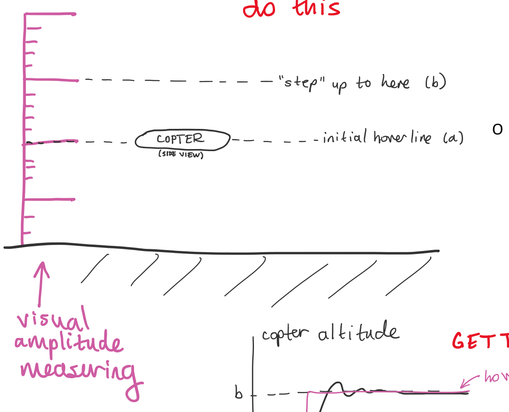
Great Controls Labs Ideas

Friday, August 22, 2025 12:44 PM

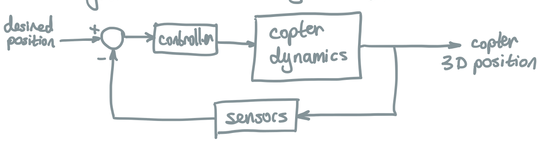
Ideas for great labs

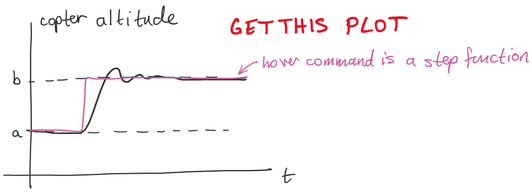
- Hover
  - What sensors for this? Probably something to measure altitude as well as a camera or something to find a physical marker on the ground marking where to hover
  - A PID controller would be just fine for hovering (I assume)
- Measure a real step response ☆old labs do this

input: hover command
output: altitude measurement

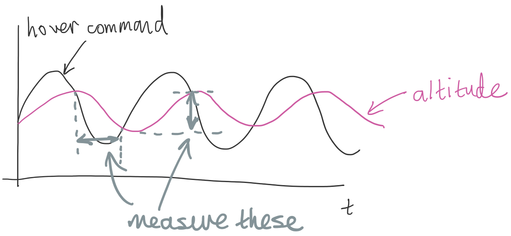


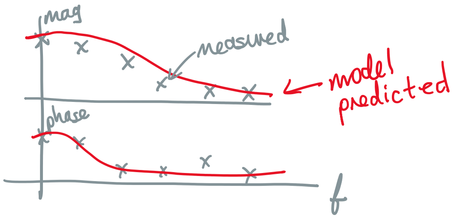
System is something like this:





- Hover with disturbances
  - External disturbances will be things like wind or things like getting hit
  - The disturbances should be repeatable if possible. Ideas:
    - Fan, should be movable to provide a disturbance in various directions but also able to be set still so everyone gets the same treatment
    - Bopping the copter with your hand--not very repeatable but very easy to do
    - Bopping or torquing the copter with some mechanism that does the same thing every time?
  - Could repeat the step response test in the presence of disturbance and compare the results to the undisturbed case
- Measure the hover-copter's frequency response physically ☆old labs do this
  - Make the "where to hover" command a sinusoid
  - Measure the amplitude of the resulting up-and-down motion of the copter (magnitude response)
    - This could maybe done physically/visually, which I think would help the students understand what we are measuring.
  - Measure the phase shift between the altitude of the copter and the input signal (phase response)
    - I can't imagine how to do this visually. Ideas?
    - Either way we will need some kind of oscope-like data visualizer tool. It could be for post processing since the thing will be flying.
  - Repeat this ~8 times with different input frequencies until you can approximate a Bode plot/freq domain plot (analogous to the CTFT)





- Compare measured results to the results predicted by a model
  - This means we need to choose a base system that HAS a reasonable (maybe linearized) transfer function/block diagram model that's close enough to be interesting
  - I think there exist such things for quadcopters, but I should go check.
  - This also means we need to maybe write some pretty python to help visualize data and run simulations from models and compare the two.
  - Currently, I give them Matlab scripts that make nice plots and they just have to put their data and measured model parameters in one line of the code.
- Investigate a wide variety of controller performance measures and learn how to measure/quantify each of these (and possibly others) for this real system. This could (and should) be a design competition: *who can make the best hover controller?*
  - Stability and stability margins (gain margin, phase margin)
  - Disturbance rejection
  - Percent overshoot
  - Settling time
  - Steady state error
  - Total control effort
- Tracking something
  - This is harder than hovering, but our system should support it.
  - Maybe a remote control car with a special visual marker on top, and you have to make your copter follow it?
  - There are ways to extend some of the above stuff to this scenario, but I think that if we did all of the above analysis stuff and also made the copters track a path or a moving goal on the ground, that would be an amazing lab semester.
  - Need to think of ways to make this into a fun competition as well